R Reference Sheet

Tips:

1. When in the console, the up arrow retrieves your previous command.
2. You can auto-complete by hitting the tab key.
3. We always need to specify the path to a file unless it is in the working directory. We can also specify the path from the working directory to the file.
4. Typing `Control+Enter` on Windows or `Command+Return` on Mac will run the line of code your cursor is on.
5. Typing `Option+minus` will enter the arrow operator `<-` with spaces around it.
6. Typing `Control+Shift+M` on Windows or `Command+Shift+M` gives the pipe (`|>` or `%>%`).
7. In **R**, the comment symbol is `#`.
8. Type a question mark in front of a function or data set to get documentation about it. Example: `?rep`.

Important Base R Functions

Function	Explanation
`c()`	Creates a vector.
`data.frame()`	Creates a data frame.
`install.packages("")`	Installs the indicated package. The name must be in quotes. Only need to use once per package.
`library()`	Loads a library/package to give access to its functions and data sets.
`read.csv("")`	Reads in a comma separated value file. Put file name in quotes.
`read.table("")`	Reads in a text file. Put file name in quotes.
`factor()`	Turns a vector into a factor (categorical variable).
`length()`	Counts the number of elements in a vector.
`rep()`	Repeat a value, string, or vector.
`mean()`	Computes the average of a vector.
`sd()`	Computes the standard deviation of a vector.
`median()`	Computes the median of a vector.
`quantile()`	Computes a quantile. Put 0.25 as second entry for Q_1, 0.75 for Q_3.
`min()`	Finds the minimum of a vector.
`max()`	Finds the maximum of a vector.
`sum()`	Adds up all values in a vector.
`table()`	Counts how many of each category or value for a vector.
`summary()`	Computes the 5-number summary (and the mean) of a vector.
`round()`	Rounds a number or vector to the desired number of decimal places.
`data()`	Loads a data set into memory.
`class()`	Find the class an object belongs to, like vector, factor, data frame, etc.
`exp()`	Compute the number e raised to a power.
`log()`	Compute the log of a number (natural log by default).
`unique()`	Returns all of the different (unique) values in a vector.

`cbind()`	Binds columns.
`rbind()`	Binds rows.
`nrow()`	Gives number of rows in data frame.
`ncol()`	Gives number of columns in data frame.
`dim()`	Gives number of rows and columns in data frame.
`rm()`	Removes a variable from the environment.
`as.factor()`	Changes the type of a vector to a factor.
`as.character()`	Changes the type of a vector to a character.
`as.numberic()`	Changes the type of a vector to numeric.
`na.omit()`	Removes all NA values from a vector or all NA rows from a data frame.
`function(){ }`	Creates a function. The function body should be between { and }.

Assigning and Subsetting Vectors

Command	Result	Explanation
`x <- c(2, 4, 6, 8)`	2 4 6 8	Join elements into a vector.
`x = c(2, 4, 6, 8)`	2 4 6 8	Alternative way to join elements into a vector.
`2:6`	2 3 4 5 6	An integer sequence.
`seq(2, 3, by = 0.5)`	2.0 2.5 3.0	A more complicated sequence.
`rep(1:2, times = 3)`	1 2 1 2 1 2	Repeat a vector.
`rep(1:2, each = 3)`	1 1 1 2 2 2	Repeat elements of a vector.
`x[4]`	8	Take the fourth element of `x`.
`x[-4]`	2 4 6	Take all but the fourth element of `x`.
`x[2:4]`	4 6 8	Take elements two through four of `x`.
`x[c(1, 3)]`	2 6	Take elements one and three of `x`.
`x[-c(2, 3)]`	2 8	Take all but the second and third elements of `x`.
`x[x == 6]`	6	Take elements that are equal to 10 in `x`.
`x[x < 5]`	2 4	Take elements that are less than 5 in `x`.
`x[x %in% c(1, 2, 3, 4)]`	2 4	Take elements in the set 1, 2, 3, 4 in `x`.
`x["apple"]`		Take elements in `x` named "apple".

Assigning and Subsetting Data Frames or Tibbles

Command	Explanation
`df <- data.frame(`	
`  x = 1:3,`	Creates data frame with columns `x` and `y`.
`  y = c("a", "b", "c")`	An equal sign can also be used instead of the arrow.
`)`	
`df$x`	Takes the `x` column in the data frame.
`df$x[1:3]`	Takes the 1st through 3rd element the `x` column.
`df[, 2]`	Takes everything in the 2nd column of the data frame.
`df[3, ]`	Takes everything in the 3rd row of the data frame.
`df[2:3, 1]`	Takes the 2nd and 3rd rows in the first column.
`df[[1]]`	Takes the first column.

Important Tidyverse Functions

Type `library(tidyverse)` after you've installed it with `install.packages("tidyverse")` to load the tidyverse and have access to these functions.

Function	Explanation
`tibble()`	Creates a tibble, which is a more modern data frame.
`mutate(data, variable)`	Creates a new variable in the given data frame.
`filter(data, condition)`	Subsets data frame based on the condition. Selects rows.
`select(data, variables)`	Selects variables based on their name. Selects columns.
`summarize(data, functions)`	Computes the function value for a variable in the data frame.
`arrange(data, variable)`	Changes the ordering of the rows. The `desc()` function is useful.
`group_by(data, variable)`	allows you to perform any operation "by group". Useful before using `summarize()`.
`pivot_longer(data, cols)`	Changes data format from not-tidy to tidy.
`read_csv("")`	Reads in a csv file as a tibble. Put file name in quotes.
`read_table("")`	Reads in a text file as a tibble. Put file name in quotes.
`read_excel("")`	Read in excel file. Put file name in quotes. Requires `readxl` package.

Important ggplot2 Functions

`ggplot2` is part of the tidyverse, so typing `library(tidyverse)` will also load `ggplot2`.

Every ggplot needs three components:

1. The data in tidy format (one observation/data value per row with variables in each column).
2. An aesthetic mapping using the `aes()` function, which usually goes in the `ggplot()` function.
3. A geometry using a `geom_something()` functon.

Function	Explanation
`ggplot(data, aes())`	The core plotting function. Needs a data frame/tibble entered and often has the `aes()` function if the arguments of `aes()` apply to all.
`aes()`	The aesthetic map. Typical arguments include `x`, `y`, `fill`, `color`, `shape`.
`geom_point()`	Creates a scatter plot.
`geom_bar()`	Creates a bar plot.
`geom_boxplot()`	Creates a box plot.
`geom_histogram()`	Creates a histogram.
`geom_density()`	Creates a density plot.
`geom_abline()`	Adds a line to the plot with the `slope` and `intercept` arguments.
`geom_qq()`	Creates a quantile-quantile (QQ) plot.
`geom_qq_line()`	Put a QQ line on a QQ plot.
`labs()`	Changes the labels for `x`, `y`, `title`, `fill`, `color`, `shape`, etc.
`lims()`	Changes the limits on `x`, or `y` or both.
`theme()`	Allows for the adjustments of many options on the graph, like text size.
`theme_bw()`	One of several `theme_` functions that changes the plot style. Put this function before the `theme()` function if both are used.

Each of the above functions, with the exception of `ggplot()` and `aes()`, is added on to the previous function with a plus sign (`+`). An example of some ggplot code is given below:

```
ggplot(mtcars, aes(x = wt, y = mpg, color = factor(cyl))) +
  geom_point(size = 2) +
  labs(x = "Weight", y = "MPG", color = "Cylinders", title = "ggplot!") +
  theme(axis.title = element_text(size = 14),
        plot.title = element_text(size = 18, face = "bold"),
        legend.position = "bottom")
```

Probability Functions

Function	Explanation
`dbinom()`	Finds $P(X = x)$ for a binomial.
`pbinom()`	Finds $P(X \leq x)$ for a binomial.
`rbinom()`	Randomly generates values from a binomial distribution.
`dpois()`	Finds $P(X = x)$ for a Poisson.
`ppois()`	Finds $P(X \leq x)$ for a Poisson.
`rpois()`	Randomly generates values from a Poisson distribution.
`pnorm()`	Finds $P(X \leq x) = P(X < x)$ for a normal.
`qnorm()`	Finds a percentile for a normal.
`rnorm()`	Randomly generates values from a normal distribution.

R also has functions that begin with `p`, `q`, and `r` for `t`, `chisq`, `f` that work the same way as the `pnorm()`, `qnorm()`, and `rnorm()` functions.

More Advanced Statistics Functions

Function	Explanation
`cor(x, y)`	Computes the correlation between two vectors.
`t.test()`	Performs a t test for one or two means.
`prop.test()`	Performs a test for one or two proportions.
`chisq.test()`	Performs a goodness-of-fit test or test for independence.
`cor.test()`	Performs a correlation test.
`lm(y ~ x, data = df)`	Creates a linear model.
`glm(y ~ x, data, family)`	Creates a generalized linear model.
`aov(y ~ x, data = df)`	Creates an ANOVA object.
`anova(model)`	Outputs an ANOVA table for a given model.
`predict()`	Obtain predictions, confidence intervals, and/or prediction intervals.
`prcomp()`	Used for principal component analysis.
`wilcox.test()`	Performs a Wilcoxon rank sum test.
`kruskal.test()`	Performs a Kruskal-Wallis test.